

Chapters 16 & 17

Electrochemistry

Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapter 16 Problems:

16-7, 16-8, 16-9, 16-10, 16-13, 16-16, 16-20, 16-22

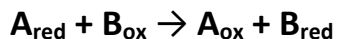
Chapter 17 Problems:

17-8, 17-9, 17-11

Table 19.1 Standard Electrode (Reduction) Potentials in Aqueous Solution at 25°C*

Cathode (Reduction) Half-Reaction	Standard Potential, E° (Volts)
$\text{Li}^+(aq) + e^- \rightleftharpoons \text{Li}(s)$	-3.04
$\text{Na}^+(aq) + e^- \rightleftharpoons \text{Na}(s)$	-2.71
$\text{Mg}^{2+}(aq) + 2e^- \rightleftharpoons \text{Mg}(s)$	-2.38
$\text{Al}^{3+}(aq) + 3e^- \rightleftharpoons \text{Al}(s)$	-1.66
$2\text{H}_2\text{O}(l) + 2e^- \rightleftharpoons \text{H}_2(g) + 2\text{OH}^-(aq)$	-0.83
$\text{Zn}^{2+}(aq) + 2e^- \rightleftharpoons \text{Zn}(s)$	-0.76
$\text{Cr}^{3+}(aq) + 3e^- \rightleftharpoons \text{Cr}(s)$	-0.74
$\text{Fe}^{2+}(aq) + 2e^- \rightleftharpoons \text{Fe}(s)$	-0.41
$\text{Cd}^{2+}(aq) + 2e^- \rightleftharpoons \text{Cd}(s)$	-0.40
$\text{Ni}^{2+}(aq) + 2e^- \rightleftharpoons \text{Ni}(s)$	-0.23
$\text{Sn}^{2+}(aq) + 2e^- \rightleftharpoons \text{Sn}(s)$	-0.14
$\text{Pb}^{2+}(aq) + 2e^- \rightleftharpoons \text{Pb}(s)$	-0.13
$\text{Fe}^{3+}(aq) + 3e^- \rightleftharpoons \text{Fe}(s)$	-0.04
$2\text{H}^+(aq) + 2e^- \rightleftharpoons \text{H}_2(g)$	0.00
$\text{Sn}^{4+}(aq) + 2e^- \rightleftharpoons \text{Sn}^{2+}(aq)$	0.15
$\text{Cu}^{2+}(aq) + e^- \rightleftharpoons \text{Cu}^+(aq)$	0.16
$\text{Cu}^{2+}(aq) + 2e^- \rightleftharpoons \text{Cu}(s)$	0.34
$\text{IO}^-(aq) + \text{H}_2\text{O}(l) + 2e^- \rightleftharpoons \text{I}^-(aq) + 2\text{OH}^-(aq)$	0.49
$\text{Cu}^+(aq) + e^- \rightleftharpoons \text{Cu}(s)$	0.52
$\text{I}_2(s) + 2e^- \rightleftharpoons 2\text{I}^-(aq)$	0.54
$\text{Fe}^{3+}(aq) + e^- \rightleftharpoons \text{Fe}^{2+}(aq)$	0.77
$\text{Hg}_2^{2+}(aq) + 2e^- \rightleftharpoons 2\text{Hg}(l)$	0.80
$\text{Ag}^+(aq) + e^- \rightleftharpoons \text{Ag}(s)$	0.80
$\text{Hg}_2^{2+}(aq) + 2e^- \rightleftharpoons \text{Hg}(l)$	0.85
$\text{ClO}^-(aq) + \text{H}_2\text{O}(l) + 2e^- \rightleftharpoons \text{Cl}^-(aq) + 2\text{OH}^-(aq)$	0.90
$2\text{Hg}_2^{2+}(aq) + 2e^- \rightleftharpoons \text{Hg}_2^{2+}(aq)$	0.90
$\text{NO}_3^-(aq) + 4\text{H}^+(aq) + 3e^- \rightleftharpoons \text{NO}(g) + 2\text{H}_2\text{O}(l)$	0.96
$\text{Br}_2(l) + 2e^- \rightleftharpoons 2\text{Br}^-(aq)$	1.07
$\text{O}_2(g) + 4\text{H}^+(aq) + 4e^- \rightleftharpoons 2\text{H}_2\text{O}(l)$	1.23
$\text{Cr}_2\text{O}_7^{2-}(aq) + 14\text{H}^+(aq) + 6e^- \rightleftharpoons 2\text{Cr}^{3+}(aq) + 7\text{H}_2\text{O}(l)$	1.33
$\text{Cl}_2(g) + 2e^- \rightleftharpoons 2\text{Cl}^-(aq)$	1.36
$\text{MnO}_4^-(aq) + 8\text{H}^+(aq) + 5e^- \rightleftharpoons \text{Mn}^{2+}(aq) + 4\text{H}_2\text{O}(l)$	1.49
$\text{H}_2\text{O}_2(aq) + 2\text{H}^+(aq) + 2e^- \rightleftharpoons 2\text{H}_2\text{O}(l)$	1.78
$\text{S}_2\text{O}_8^{2-}(aq) + 2e^- \rightleftharpoons 2\text{SO}_4^{2-}(aq)$	2.01
$\text{F}_2(g) + 2e^- \rightleftharpoons 2\text{F}^-(aq)$	2.87

Introduction to Electrochemistry



Rules for assigning oxidation states (in order of priority)

1. free elements have an oxidation state = 0
2. monatomic ions have an oxidation state equal to their charge
3. (a) the sum of the oxidation states of all the atoms in a compound is 0
(b) the sum of the oxidation states of all the atoms in a polyatomic ion equals the charge on the ion
4. (a) Group I metals have an oxidation state of +1 in all their compounds
(b) Group II metals have an oxidation state of +2 in all their compounds
5. in their compounds, nonmetals have oxidation states according to the table below
 - nonmetals higher on the table take priority

Nonmetal	Oxidation State	Example
F	-1	CF ₄
H	+1	CH ₄
O	-2	CO ₂
Group 7A	-1	CCl ₄
Group 6A	-2	CS ₂
Group 5A	-3	NH ₃

Redox Reactions

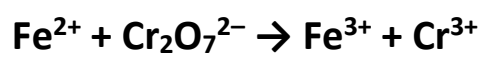
Reactions where electrons are transferred from one atom to another are called oxidation–reduction reactions

Atoms that lose electrons are being oxidized, atoms that gain electrons are being reduced

Balancing Redox Reactions

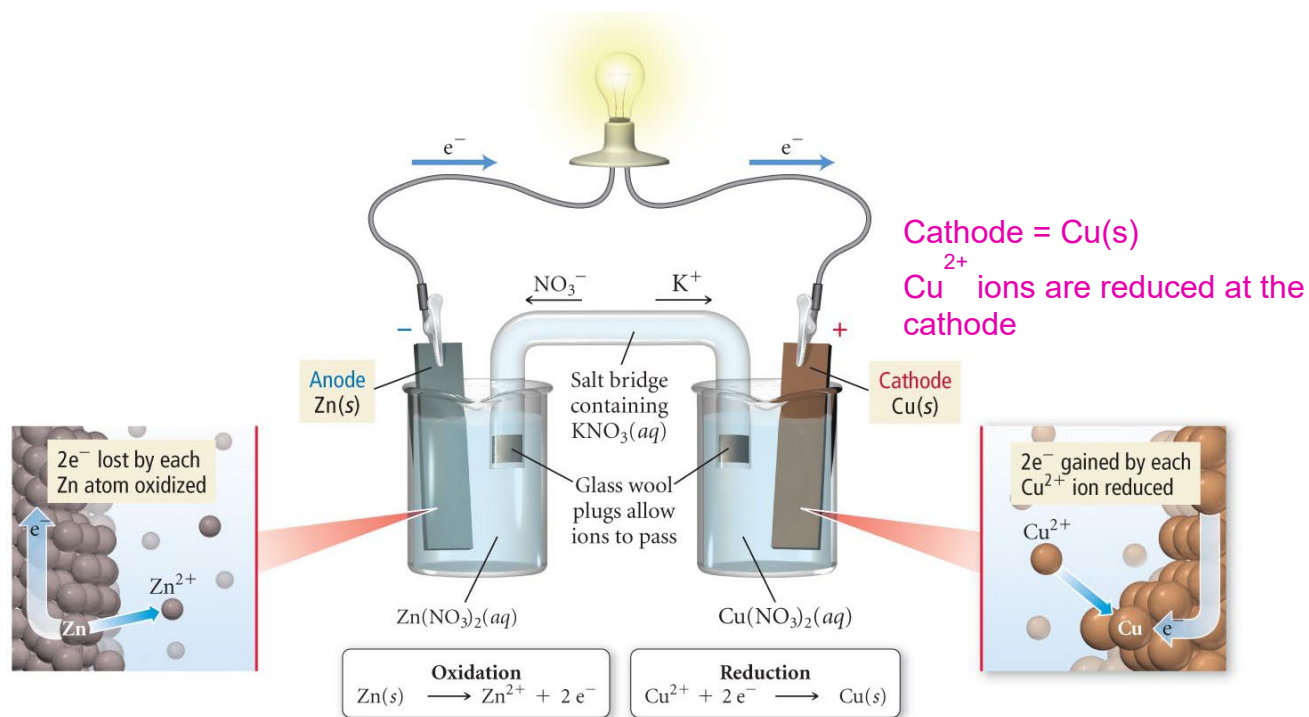
- 1. assign oxidation states**
 - a) determine element oxidized and element reduced**
- 2. write ox. & red. half-reactions, including electrons**
 - a) ox. electrons on right, red. electrons on left of arrow**
- 3. balance half-reactions by mass**
 - a) first balance elements other than H and O**
 - b) add H₂O where need O**
 - c) add H⁺ where need H**
 - d) if reaction done in Base, neutralize H⁺ with OH⁻**
- 4. balance half-reactions by charge**
 - a) balance charge by adjusting electrons**
- 5. balance electrons between half-reactions**
- 6. add half-reactions**
- 7. check by counting atoms and total charge**

Balance for the following redox reactions



Oxidizing agent is reduced (where the electrons go)

Reducing agent is oxidized (source of electrons)



anode || cathode

$\text{Zn}(\text{s}) \mid \text{Zn}^{2+}(\text{aq}) \parallel \text{Cu}^{2+}(\text{aq}) \mid \text{Cu}(\text{s})$

The difference in potential energy between the reactants and products is the potential difference (units = Volt J/C)

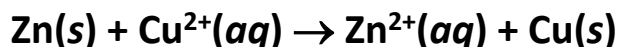
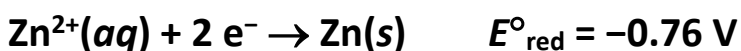
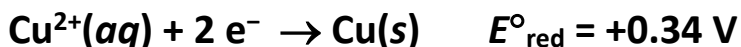
The difference in potential energy between the anode and the cathode in a voltaic cell is called the cell potential (electromotive force = emf)

The cell potential under standard conditions is called the standard emf, E°_{cell}

25 °C, 1 atm for gases, 1 M concentration of solution

Sum of the cell potentials for the half-reactions

- $E^\circ_{\text{cell}} = E^\circ_{\text{reduction}} + E^\circ_{\text{oxidation}}$
- $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ **(in terms of all reduction reactions)**
- When adding E° values for the half-cells, *do not multiply* the half-cell E° values, even if you need to multiply the half-reactions to balance the equation

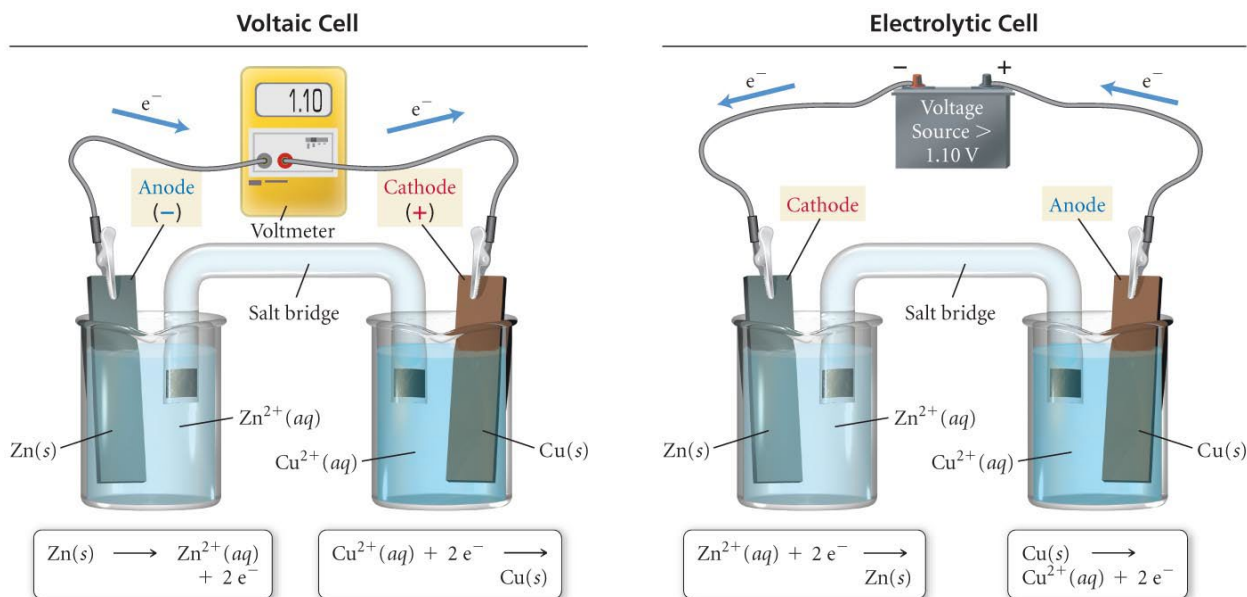


$$E^\circ_{\text{cell}} = 0.34 - (-0.76) = 1.10 \text{ V (Spontaneous)}$$

Galvanic (aka Voltaic) cells have $+E^\circ_{\text{cell}}$ (batteries)

These redox reactions occur spontaneously (ΔG is neg)

aka Galvanic



© 2011 Pearson Education, Inc.

a

Electrolytic cells requires an external source of electricity (charging a battery)

We use electrical energy to overcome the energy barrier of a non-spontaneous reaction, allowing it to occur

The reaction that takes place is the opposite of a spontaneous process

Galvanic Cells Store Energy

Electrolytic Cells Consume Energy

Regardless of whether it is a Galvanic or Electrolytic cell oxidation always occurs at the anode and reduction always occurs at the cathode

Currents in Electrochemical Cells

- 1. Electrons carry the charge within the electrodes as well as the external conductor**
- 2. Anions and cations are the charge carriers with in the cell**
- 3. The ionic conduction of the solution is coupled to the electronic conduction in the electrodes by the reduction reaction at the cathode and the oxidation reaction at the anode through the salt bridge.**

Electrode Potentials

The potential difference between electrodes of a cell is a measure of the tendency for the reaction to occur

For a spontaneous reaction

one that proceeds in the forward direction with the chemicals in their standard states

$$\Delta G^\circ < 0 \text{ (negative)}$$

$E^\circ > 0$ (positive) (go to standard reduction potential table find best anode and cathode)

$$K > 1$$

$$\Delta G^\circ = -RT \ln K = -nFE^\circ_{\text{cell}}$$

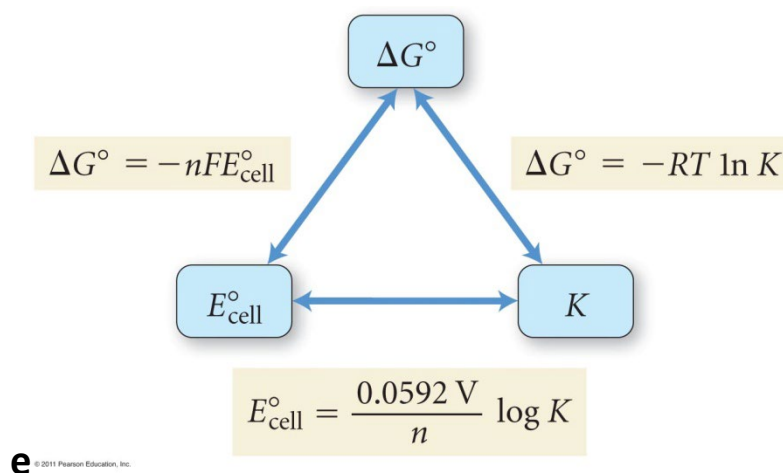
R = molar gas constant 8.314 J/mol K

T = temperature in K

n is the number moles of electrons

F = Faraday's Constant = 96,500 C/mol e

ln = 2.303 log



$$\Delta G = \Delta G^\circ + RT \ln Q$$

(Q= reaction quotient)

$$-nFE = -nFE^\circ + RT \ln Q$$

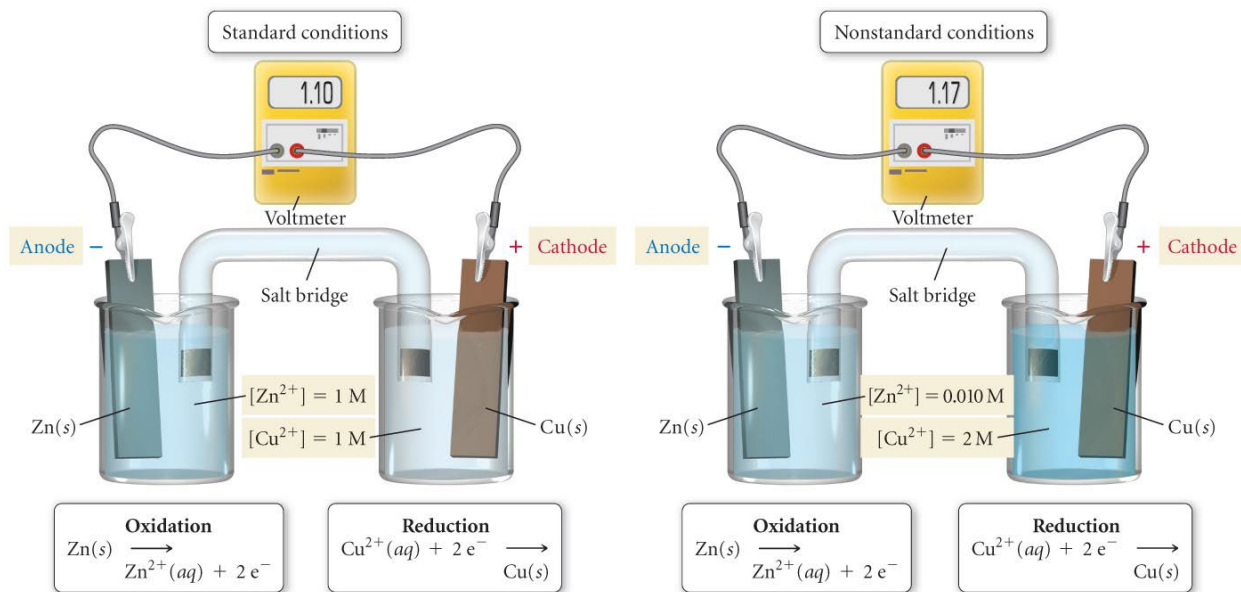
$$\frac{-nFE}{-nF} = \frac{-nFE^\circ}{-nF} + \frac{RT \ln Q}{-nF}$$

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

When the temperature is 25 °C

$$E = E^\circ - \frac{0.0592}{n} \log Q$$

E_{cell} When Ion Concentrations are Not 1 M



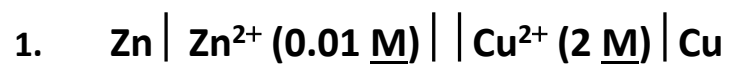
Consider the reversible redox reaction



$$E = E^\circ - \frac{RT}{nF} \ln \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$$

Nernst Equation

$$E = E^\circ - \frac{0.0592}{n} \log \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$$

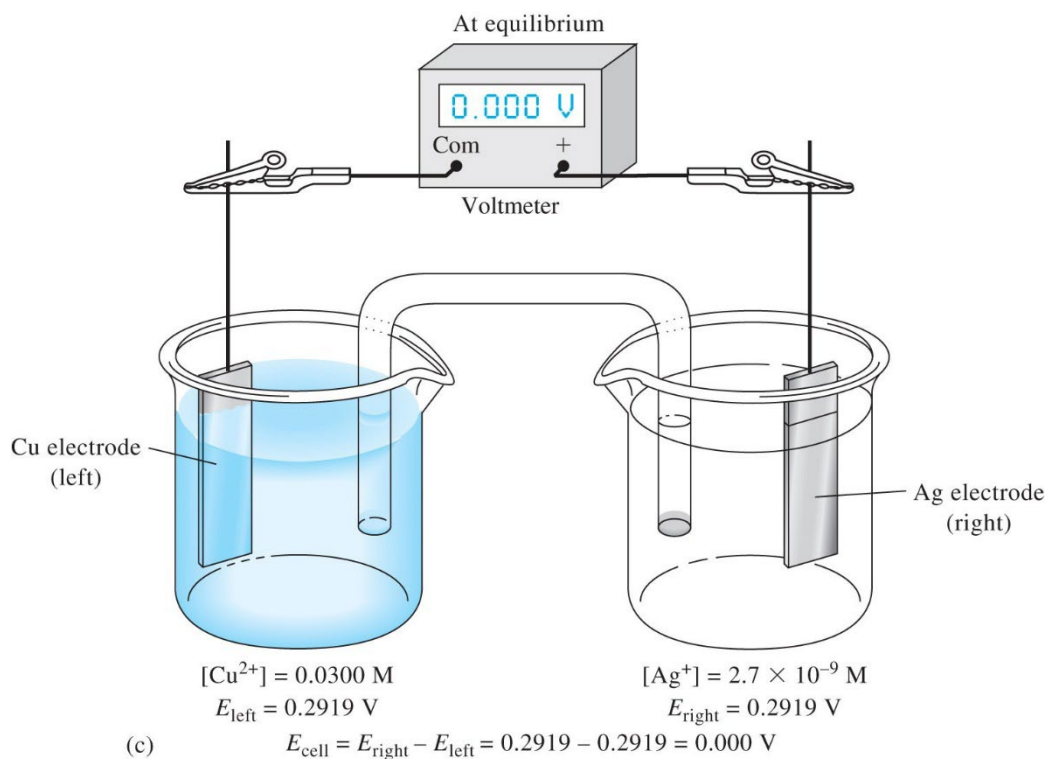


Write the balanced overall reaction

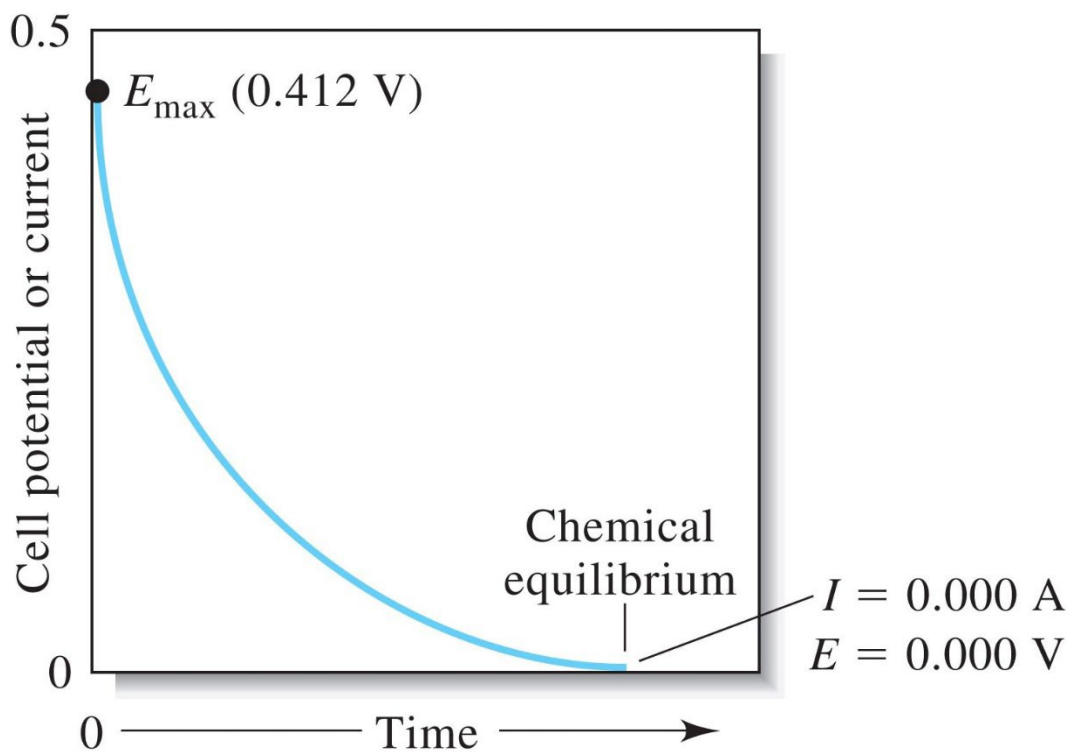
Calculate E° cell

Calculate E cell

Determine if it is galvanic or electrolytic



© Cengage Learning. All Rights Reserved.



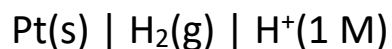
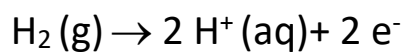
© Cengage Learning. All Rights Reserved.

Standard Hydrogen Reference Electrode

We cannot measure the absolute tendency of a half-reaction; we can only measure it relative to another half-reaction

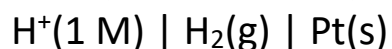
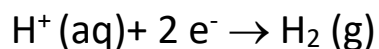
We select as a standard half-reaction the reduction of H^+ to H_2 under standard conditions, which we assign a potential difference = 0 v

Anode

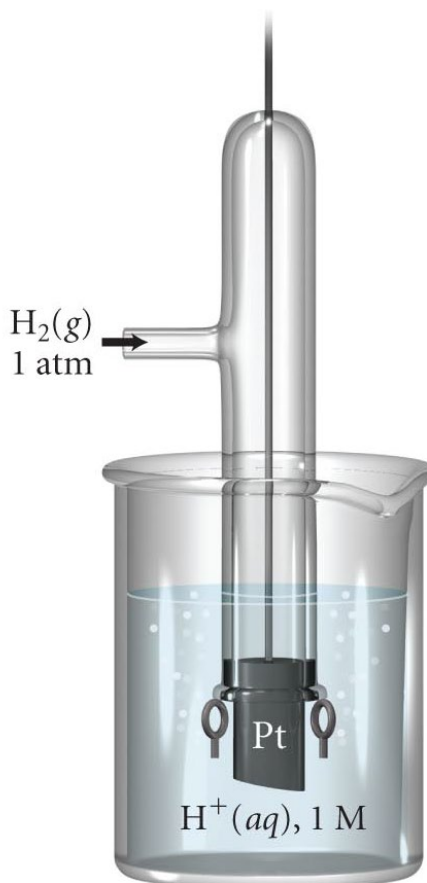


SHE

Cathode



SHE

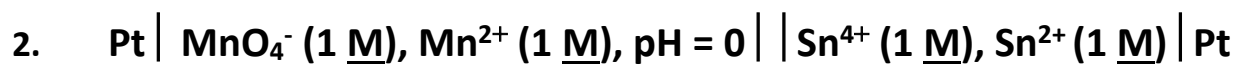


The Standard Electrode Potential E°

- 1. Is relative in the sense that it is the potential of an electrochemical cell in which the left-hand side is the SHE**
- 2. Refers exclusively to a reduction reaction**
- 3. Measures the relative force tending to drive the $\frac{1}{2}$ rxn from a state where the reactants and products are at unit activity to a state in which the reactants and products are at their equilibrium activities relative to the SHE**
- 4. It is independent of the number of moles of reactants and product in the balanced $\frac{1}{2}$ rxn**

5. A positive electrode potential indicates that the half-reaction in question is spontaneous with respect to the SHE (the oxidant in the $\frac{1}{2}$ rxn is a stronger oxidant than the H^+)
6. It is temperature dependent

Typical half-cell reactions



Write the balanced overall reaction

Calculate E° cell

Determine if it is galvanic or electrolytic

3. $\text{Pt} \mid \text{MnO}_4^- (0.250 \text{ M}), \text{Mn}^{2+} (0.191 \text{ M}), \text{pH} = 4.300 \mid \mid \text{Sn}^{4+} (0.470 \text{ M}), \text{Sn}^{2+} (0.0922 \text{ M}) \mid \text{Pt}$
Calculate E cell

There are two ways to solve this problem

It is not necessary to write out the overall balanced redox reaction

$$\Delta G^\circ = -RT \ln K_{eq}$$

$$\Delta G^\circ = -nFE^\circ_{cell}$$

$$\ln K_{eq} = -\frac{\Delta G^\circ}{RT} = -\left(\frac{-nFE^\circ_{cell}}{RT}\right)$$

$$\log K_{eq} = \frac{nE^\circ_{cell}}{0.0592}$$

$$E^\circ_{cell} = -1.34 \text{ V (from example 2)}$$

Calculate K_{eq}

Reversible cells: can be made to run backwards

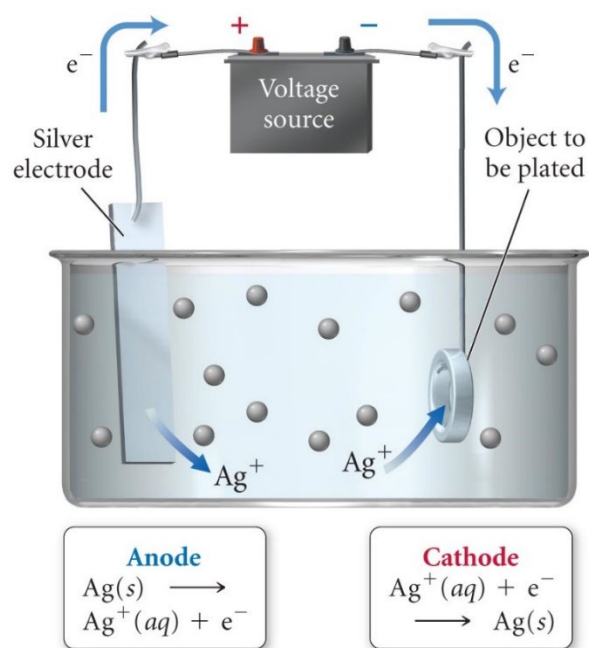
Non-reversible cells: reversing the potential does not cause a general reversal, but usually causes generation of different chemical species than were originally present

Electroplating

In electroplating, the work piece is the cathode

Cations are reduced at cathode and plate to the surface of the work piece

The anode is made of the plate metal. The anode oxidizes and replaces the metal cations in the solution.



© 2011 Pearson Education, Inc.

- When more than one cation is present, the cation that is easiest to reduce will be reduced first at the cathode
 - ✓ least negative or most positive E°_{red}
- When more than one anion is present, the anion that is easiest to oxidize will be oxidized first at the anode
 - ✓ least negative or most positive E°_{ox}

- Possible cathode reactions
 - ✓ reduction of cation to metal
 - ✓ reduction of water to H_2
- Possible anode reactions
 - ✓ oxidation of anion to element
 - ✓ oxidation of H_2O to O_2
 - ✓ oxidation of electrode
 - particularly Cu
 - graphite doesn't oxidize
- Half-reactions that lead to least negative E_{cell} will occur
 - ✓ unless overvoltage changes the conditions

Stoichiometry of Electrolysis

- In an electrolytic cell, the amount of product made is related to the number of electrons transferred
 - ✓ essentially, the electrons are a reactant
- The number of moles of electrons that flow through the electrolytic cell depends on the current and length of time
 - ✓ 1 Amp = 1 Coulomb of charge/second
 - ✓ 1 mole of e^- = 96,500 Coulombs of charge
 - Faraday's constant

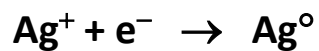
$$C = A \times s$$

$$\text{Coulombs} = \text{Amps} * \text{seconds}$$

$$C = A * s$$

$$C/s = A$$

4. A constant current deposits 365 mg of Ag^0 at the cathode of an electrolytic cell in 216 minutes from an AgNO_3 solution. What is the required current for this to occur?



5. How many grams of Pt can be deposited from a solution of PtCl_4^{2-} if 200.0 A of current flows for 7 days?

We can also use electrochemical cells to calculate K_{sp} values

6. From the standard potentials calculate the K_{sp} for $TlCl_{(s)}$